



PRACTICE EXAMINATION

Mathematical Methods

2026

Question booklet 1

- Questions 1 to 6 (50 marks)
- Answer **all** questions
- Write your answers in this question booklet
- You may write on page 12 if you need more space
- Allow approximately 65 minutes
- Approved calculators may be used — complete the box below

Examination information

Materials

- Question booklet 1
- Question booklet 2
- Formula sheet
- Student name label

Instructions

- Show appropriate working and steps of logic in the question booklets
- State all answers correct to three significant figures, unless otherwise instructed
- Use black or blue pen
- You may use a sharp dark pencil for diagrams and graphical representations

Total time: 130 minutes

Total marks: 100 (across both question booklets)

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Attach your student name label here

Graphics calculator

1. Brand _____
Model _____
2. Brand _____
Model _____

Question 1 (8 marks)

(a) Determine $\frac{dy}{dx}$ for the following functions. You do not need to simplify your answers.

(i) $y = 3 \cos(4x) - \ln(x^3)$.

(2 marks)

(ii) $y = x^3 \ln(2x)$.

(2 marks)

(iii) $y = \frac{5x^2 - 1}{e^{2x}}$.

(2 marks)

(b) Determine $\int \left(e^{4x} + \frac{3}{2x+1} \right) dx, x > -\frac{1}{2}.$

(2 marks)

Question 2 (9 marks)

An electric bus travels between two depots. The time taken to complete the trip, T minutes, is normally distributed with a mean of $\mu = 42$ minutes and a standard deviation of $\sigma = 3.5$ minutes.

(a) Determine

(i) $\Pr(T < 40)$.

(1 mark)

(ii) $\Pr(38 \leq T \leq 45)$.

(1 mark)

(b) The timetable allows k minutes for the trip. Determine the value of k for which 95% of trips are completed within the time allowed.

(2 marks)

(c) Let $\bar{T}_6$ be the random variable representing the sample mean of the times of 6 randomly selected trips.

(i) State the mean and the standard deviation of $\overline{T}_6$.

(2 marks)

(d) A student states that $\Pr(\bar{T}_6 > 44)$ should be equal to $\Pr(T > 44)$. Explain why the student is incorrect.

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- (c) Let $h(t)$ be a function that models the height of the drone above the landing pad, in metres, t seconds after it is released.

Show that $h(t) = 12t + 24e^{-0.5t} - 24$. Assume that $h(0) = 0$.

(3 marks)

- (d) Determine the height of the drone 10 seconds after it is released.

(1 mark)

- (e) Determine how long after its release the drone first reaches a height of 50 metres.

(2 marks)

Consider the function $f(x) = \frac{1}{2x+1}$, where $x \neq -\frac{1}{2}$.

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A vending machine at a train station gives a random number of bonus tokens with each purchase. Let X represent the number of bonus tokens given with a randomly selected purchase. The probability distribution of X is shown in Table 1.

x	0	1	2	3
$\Pr(X = x)$	0.45	0.30	a	0.05

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- (i) Fill in the boxes below to show the value of the parameters of Y .

(1 mark)

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Question 6 (9 marks)

A council surveys a random sample of 250 residents about a proposed bike lane. Of the residents surveyed, 145 support the proposal.

- (a) Determine the sample proportion of residents who support the proposal.

(1 mark)

- (b) Fill in the boxes below to form a 95% confidence interval for the population proportion of residents who support the proposal, p .

$$\boxed{} \leq p \leq \boxed{}$$

(2 marks)

- (c) The council will proceed with the proposal only if it can be confident that *more than half* of all residents support it.

- (i) Does the confidence interval calculated in part (b) support proceeding with the proposal, with 95% confidence? Tick the appropriate box to indicate your answer.

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Yes

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No

(1 mark)

- (ii) Justify your answer to part (c)(i).

(1 mark)

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